

Mapping artificial intelligence in marketing: insights through a two-decade analysis (2000–2024)

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Mapeado bibliométrico de la Inteligencia Artificial en Marketing: Perspectivas a través de un análisis de dos décadas (2000-2024)

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Abstract

Purpose – Integrating Artificial Intelligence (AI) into marketing has driven significant transformations, reshaping consumer engagement, campaign management and business strategies. Although previous reviews have examined AI in marketing, most are limited to specific applications or relatively short time frames. There is a gap in recent bibliometric analyses that capture AI research's intellectual evolution and rapid diversification in this field. This study addresses this need by mapping the development of AI in marketing from 2000 to 2024, identifying emerging trends, key contributors and persistent knowledge gaps.

Design/methodology/approach – This study conducts a bibliometric analysis of 2,105 articles published between 2000 and 2024, using the PRISMA method and drawing on Scopus and Web of Science data. Bibliometrix and VOSviewer were employed to identify the most influential journals, authors and countries and to uncover thematic clusters through co-occurrence and co-citation network analyses.

JEL Classification — M31, C38, C55, O33, D83, D91

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Findings – The results reveal an exponential growth in AI-related marketing research since 2017, with 2024 marking the highest annual publication volume. Key research areas emerging from this trend include AI ethics, generative AI applications and the strategic importance of social presence in fostering brand-consumer engagement. This study not only provides a comprehensive overview of AI's evolution and current landscape in marketing but also identifies persistent gaps, such as algorithmic bias and content authenticity, that require further investigation, thereby highlighting critical priorities for future research and practical implementation.

Originality/value – By providing the most up-to-date, systematic and longitudinal bibliometric mapping of AI research in marketing, this study substantially extends and updates previous literature reviews. It delivers a foundation for both scholars and practitioners, clarifying how the field has evolved, particularly towards generative AI and ethical considerations, and establishing a clear research agenda for addressing unresolved challenges, with specific future research avenues aligned with the latest trends.

Keywords Artificial intelligence, Marketing, Bibliometric analysis, Business evolution, Consumer behavior

Paper type Literature review

Resumen

Propósito – La integración de la Inteligencia Artificial (IA) en el marketing ha impulsado transformaciones significativas, remodelando la interacción con los consumidores, la gestión de campañas y las estrategias empresariales. Aunque revisiones previas han analizado la IA en marketing, la mayoría se limita a aplicaciones específicas o a periodos temporales relativamente cortos. Existe una brecha en la literatura reciente de análisis bibliométricos que permitan captar la evolución intelectual y la rápida diversificación de la investigación en este ámbito. Este estudio responde a esta necesidad mediante el mapeo del desarrollo de la IA en marketing entre 2000 y 2024, identificando tendencias emergentes, principales contribuyentes y vacíos persistentes de conocimiento.

Diseño/metodología/enfoque – El estudio realiza un análisis bibliométrico de 2.105 artículos publicados entre 2000 y 2024, aplicando el método PRISMA y recurriendo a datos de Scopus y Web of Science. Se utilizaron Bibliometrix y VOSviewer para identificar las revistas, autores y países más influyentes, así como, para descubrir clústeres temáticos mediante análisis de co-ocurrencia y co-citación.

Resultados – Los resultados revelan un crecimiento exponencial de la investigación sobre IA en marketing desde 2017, alcanzando 2024 el mayor volumen anual de publicaciones. Entre los temas más relevantes emergen la ética de la IA, las aplicaciones de IA generativa y la importancia estratégica de la presencia social para fortalecer el engagement marca-consumidor. El estudio no solo proporciona una visión exhaustiva de la evolución y el estado actual de la IA en marketing, sino que también identifica vacíos persistentes, como el sesgo algorítmico y la autenticidad del contenido, que requieren una investigación más profunda, señalando prioridades críticas para futuras investigaciones e implementación práctica.

Originalidad/valor – Al ofrecer el mapeo bibliométrico más actualizado, sistemático y longitudinal sobre la investigación en IA en marketing, este estudio amplía y actualiza sustancialmente las revisiones previas. Proporciona una base tanto para académicos como para profesionales, clarificando la evolución del campo, especialmente hacia la IA generativa y las consideraciones éticas, y estableciendo una agenda clara de investigación futura para abordar los desafíos no resueltos, con líneas de investigación alineadas con las tendencias más recientes.

Palabras clave Inteligencia artificial, Marketing, Análisis bibliométrico, Evolución empresarial, Comportamiento del consumidor

Tipo de papel Trabajo de investigación

1. Introduction

Artificial Intelligence (AI) is recognized as a subfield of computer science (Kumar and Suthar, 2023) that focuses on developing new systems capable of analyzing large datasets to generate future predictions (Hasan *et al.*, 2021). In this context, AI offers significant advantages to businesses while indirectly benefiting consumers (Puntoni *et al.*, 2021). AI is gaining increasing importance in social and corporate domains due to its benefits to organizations and consumers (Rivera-Montaño, 2023). However, it also presents substantial challenges resulting from the transformations it is driving (Kumar *et al.*, 2019). Innovations like machine learning, computer vision and conversational AI deliver transformative customer experiences, ranging from predictive recommendations to active participation in business improvement processes (IDC, 2021).

In the marketing domain, AI adoption is triggering profound changes across various areas (Zúñiga-Vázquez *et al.*, 2023; Gupta and Khan, 2024), generating significant interest within the scientific community (Moriuchi, 2020; Pitardi and Marriott, 2021; Fakhimi *et al.*, 2023).

This is evidenced by studies examining AI applications and their associated technologies in specific fields, such as the use of machine learning (Kumbure *et al.*, 2022), the role of AI in facilitating and optimizing digital marketing and corporate communication (Kumar *et al.*, 2020a) and AI applications in e-commerce (Bawack *et al.*, 2022). Notably, previous research, such as Vlačić *et al.* (2021), analyzed 164 articles on AI's trajectory in marketing, providing only a limited snapshot of its development. Similarly, Jain and Kumar (2024) categorized AI applications into eight thematic areas but primarily focused on conceptual groupings. While both studies provide valuable insights, they fail to offer a longitudinal perspective that captures the full breadth and complexity of AI's evolution in marketing. In this context, several key questions arise: how has research on AI applied to marketing evolved over more than two decades? What emerging topics and ethical debates define the new scientific agenda in this field? Which knowledge gaps persist and demand priority attention?

Unlike previous reviews, this study overcomes significant limitations in the literature by conducting a bibliometric analysis of research published over more than two decades. This approach enables a comprehensive and longitudinal mapping of the intellectual evolution of artificial intelligence (AI) in marketing, systematically identifying key trends, leading contributors and persistent knowledge gaps. By synthesizing a substantial body of bibliographic data, the present analysis offers a holistic perspective on how AI research in marketing has diversified and matured, directly addressing calls in recent bibliometric literature for rigorous, quantitative overviews capable of integrating large and complex datasets (Donthu *et al.*, 2021).

The central objective of this study is to map the intellectual development of AI research within the marketing domain from 2000 to 2024, highlighting emerging topics, major contributors and unresolved questions to inform future academic inquiry and professional practice. Given the unprecedented speed with which AI is transforming marketing, earlier systematic reviews quickly become obsolete and require frequent updating to reflect recent advances and novel challenges. This study delivers an exhaustive literature update, offering an up-to-date reference that complements and extends previous reviews. It thus serves as a critical resource for academics and practitioners seeking to navigate an evolving research landscape. In line with this objective, the article applies the methodological framework proposed by Mukherjee *et al.* (2022) to analyze the scientific literature on AI and its derivative technologies in marketing between 2000 and 2024, guided by the following research questions:

- RQ1. How has research on AI in the marketing field evolved, and which academic sources have exerted the greatest influence in this domain?
- RQ2. What are the main areas of inquiry and the most promising future research directions in the application of AI to marketing?
- RQ3. Which research gaps persist, and how might they be addressed in future studies?

This article is structured as follows: The next section outlines the methodology employed and the analysis results detailed in Section Three. Section four highlights the study's findings and theoretical and practical contributions, while sections five and six discuss research limitations and suggest future research directions. Finally, the conclusions drawn from the findings are presented.

2. Methodology

This study conducts a bibliometric analysis to examine the state and evolution of AI research in marketing from 2000 to 2024. This methodology provides an overview of the study published in this field based on data extracted from the Scopus and WOS academic databases (Calderón-Fajardo *et al.*, 2023; Oludapo *et al.*, 2024). Furthermore, this approach offers advantages over other literature review methods, as it is more precise, rigorous and objective. This section describes the process undertaken to collect the data analyzed in this study, following the recommendations of Donthu *et al.* (2021) (See Figure 1).

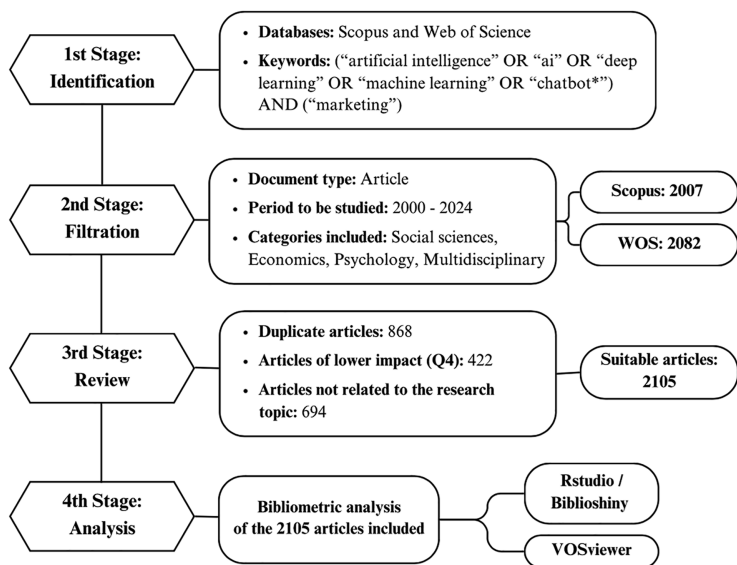


Figure 1. Stages of the bibliometric review process on AI in marketing (2000–2024). *Source:* Own elaboration

2.1 First stage: identification and selection

The WOS and Scopus databases were selected due to their significance (Rojas-Lamorena et al., 2022), global recognition (Kumar et al., 2020a, b, 2021) and successful application in previous research on marketing and market studies (Paul and Bhukya, 2021). The selected keywords were: (“artificial intelligence” OR “AI” OR “deep learning” OR “machine learning” OR “chatbot”) AND (“marketing”), applying the TS (topics) filter to include terms present in the title, abstract or keywords. This ensured the inclusion of all relevant instances where the terms mentioned earlier appear (See Figure 1). Data extraction was performed on 10 February 2025.

2.2 Second stage: filters applied

The search was refined using filters, considering only scientific articles published between 2000 and 2024 (the last complete year). Additionally, the studies had to belong to a relevant area within market research, as shown in Table 1. A total of 4,089 articles were retrieved, with 2,082 sourced from WOS and 2,007 from Scopus (see Figure 1).

2.3 Stages 3 and 4: debugging and analysis

To ensure methodological consistency and align with established guidelines for systematic literature reviews and bibliometric analyses, this study followed recent recommendations in the field regarding journal selection criteria (Donthu et al., 2021; Mukherjee et al., 2022). Specifically, we included only articles published in journals classified above the fourth quartile (Q4) of their respective subject areas at publication, as indexed in Scopus and Web of Science. This approach is consistent with procedures adopted in recent large-scale bibliometric studies and systematic reviews, which employ journal quartile metrics as a transparent, replicable and internationally recognized quality filter (Han et al., 2021; Calderón-Fajardo et al., 2023; Varea-Calero et al., 2025). As noted by Donthu et al. (2021) and Mukherjee et al. (2022), the application of these quality criteria enhances the comparability, validity and relevance of bibliometric reviews, facilitating the identification of key trends and influential contributions in the field.

Table 1. Thematic areas included in the development of the database

WOS (web of science)	Scopus
Business	Business, management and accounting
Management	Social sciences
Operations research management science	Economics, econometrics and finance
Telecommunications	Psychology
Multidisciplinary sciences	Multidisciplinary
Psychology applied	Neuroscience
Communication	
Economics	
Social science interdisciplinary	
Neurosciences	
Psychology multidisciplinary	
Business finance	
Cultural studies	
Anthropology	
Social issues	
Psychology social	
Psychology	
Ethics	
Humanities multidisciplinary	
Psychology experimental	
Robotics	
Sociology	
Behavioral sciences	
Social work	
Psychology clinical	
History of social sciences	
Source(s): Own elaboration	

To ensure the validity and consistency of the dataset used in this study, rigorous data-cleaning and filtering procedures were implemented, including the removal of duplicates, correction of erroneous entries and normalization of terms related to keywords and affiliations (Donthu *et al.*, 2021). This process ensures an accurate representation of the research domain and strengthens the reliability of the findings. Subsequently, a second screening was conducted to assess the article's focus, excluding those outside the scope of interest. As a result, a final selection of 2,105 articles was obtained. Three complementary types of analysis were conducted: (1) co-authorship analysis, (2) co-occurrence analysis of keywords and (3) citation analysis (Tanwa and Verma, 2024). To conduct these analyses, we utilized two widely recommended bibliometric tools (Arruda *et al.*, 2022). First, we employed VOSviewer for its advanced capabilities in network visualization and its effectiveness in clearly interpreting bibliometric findings (Moral-Muñoz *et al.*, 2020; Saini *et al.*, 2022; Sáez-Ortuño *et al.*, 2023). Second, we selected Biblioshiny from the R package Bibliometrix because it offers a comprehensive set of bibliometric metrics and summary outputs (Moral-Muñoz *et al.*, 2020; Arruda *et al.*, 2022). By combining these two tools, the study maximizes their complementary strengths, ensuring both the robustness of the quantitative results and the clarity of their visualization and interpretation (Moral-Muñoz *et al.*, 2020; Beloskar *et al.*, 2024).

3. Analysis

In line with the general objective of this study this section presents the results in a way that explicitly addresses each of the research questions formulated in the introduction. To address RQ1, we first analyze the annual growth in the number of publications and citations, identifying key inflection points and the expansion of the field. Furthermore, we identify the

field’s most relevant journals and authors, providing an overview of the primary sources and contributors that have shaped the development of AI research in marketing.

Figure 2 presents the annual number of published articles between 2000 and 2024 and the total citations received yearly. The data reveal an exponential growth in AI-related publications in the marketing field. Specifically, the first sixteen years reflect a slow and gradual increase in research output, with 2017 marking a turning point. A significant publication surge is observed from this year onwards, with research output becoming increasingly concentrated in recent years. Notably, 2024 recorded a total of 739 published articles, the highest number within the analyzed period.

Table 2 presents the most relevant journals regarding the number of articles published on AI in marketing. The leading journals include *Psychology and Marketing*, *Journal of Business Research* and *Marketing Science*.

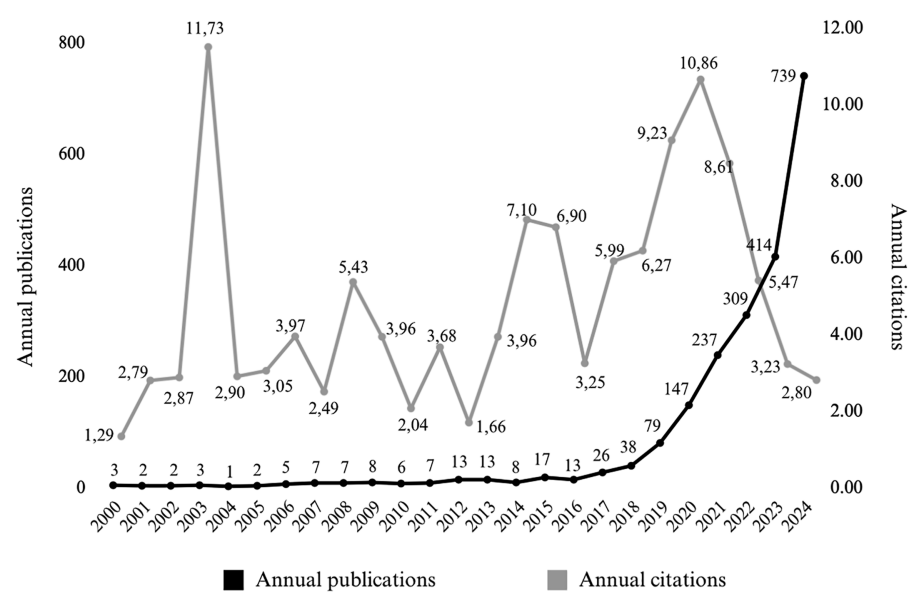


Figure 2. Annual publication and citation on AI in marketing (2000–2024). Source: Own elaboration

Table 2. The most important scientific journals on AI in marketing based on citation count (2000–2024)

Journal	Citation
Journal of Business Research	2,812
Journal of Marketing	2,763
Journal of Marketing Research	2,134
Journal of the Academy of Marketing Science	1,769
Journal of Consumer Research	1,710
Computers in Human Behavior	1,503
Marketing Science	1,440
Journal of Retailing and Consumer Service	1,300

Source(s): Own elaboration

3.1 Co-occurrence analysis (keywords)

In response to RQ2, the co-occurrence analysis identifies thematic clusters and emerging topics. It offers insight into the predominant research lines and their evolution over time and reveals the predominant thematic clusters and their interrelations within the literature. A co-occurrence analysis was conducted using VOSviewer to identify the most relevant keywords and their relationships. Based on a database of 8,971 keywords, a minimum threshold of 20 occurrences was applied, and synonyms and plurals were removed. As a result, 147 keywords were identified, forming a network comprising 135 nodes (Figure 3).

Further exploring RQ2, the thematic mapping reveals which areas drive research in the field and which themes are emerging or declining, thus informing future research priorities. Figure 3 identifies five thematic clusters:

Cluster 1 (Red): Comprising 47 keywords, with “impact” (196 occurrences and 120 links) as the most prominent term. This group focuses on consumer behavior towards AI, analyzing interactions and factors such as anthropomorphism, emotions and personality. Studies like Crolic *et al.* (2022) and Pitardi and Marriott (2021) highlight how anthropomorphic design and emotional factors influence trust in conversational agents.

Cluster 2 (Green): Composed of 30 keywords, with “machine learning” (367 occurrences and 123 links) as the dominant term. This group uses AI to explore data analysis and market research, including neural networks and advanced systems. Notable examples include Yang *et al.* (2020), who propose the Operation-aware Neural Network model, and Yun *et al.* (2021), who examine neural and behavioral responses to medical AI.

Cluster 3 (Blue): Consisting of 27 keywords, with “big data” (123 occurrences and 106 links) as the central node. This cluster focuses on integrating Big Data with AI and creating and innovating products tailored to segmented market audiences. Key studies include Kushwaha *et al.* (2021), which examines the value of Big Data in business management, and Gao and Huang (2019), which demonstrate how personalization can turn ad-exposed users into new customers.

Cluster 4 (Yellow): Containing 21 keywords, with “artificial intelligence” (573 occurrences and 133 links) as the main term. This cluster focuses on AI applied to social

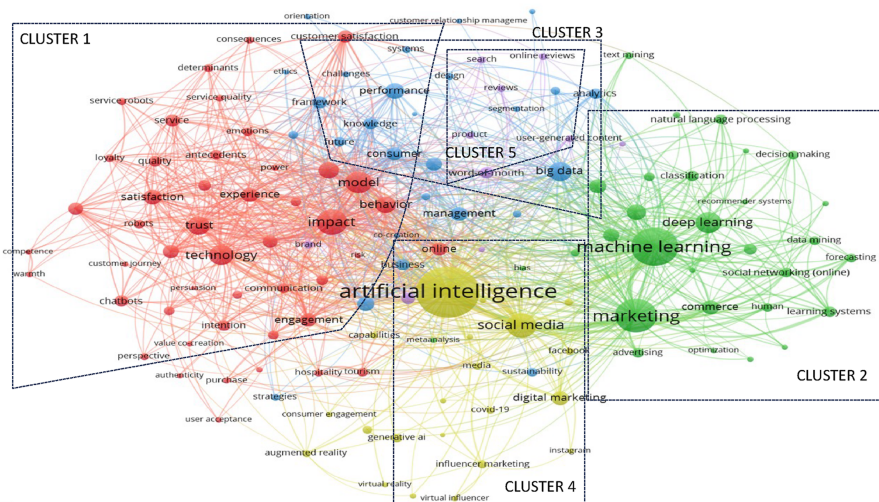


Figure 3. Co-occurrence network. Source: Own elaboration (VOSviewer)

media and influencer marketing. Noteworthy studies include [Lee et al. \(2018\)](#), which analyses the impact of social media advertising on user engagement, and [Sands et al. \(2022a, b\)](#), which explains how users interact with human and AI-generated influencers.

Cluster 5 (Purple): Comprising 10 keywords, with “word-of-mouth” (56 occurrences and 82 links) and “e-commerce” (55 occurrences and 90 links) as the key terms. Articles in this cluster examine how user-generated content and online reviews influence purchasing intent. [Li and Zhang \(2020\)](#) demonstrate the significance of online reviews in e-commerce platforms (see [Table 3](#)).

The emergence of clusters related to generative AI and ethical challenges is closely linked to recent technological advances and rising societal concerns about the trustworthiness and transparency of AI applications. These clusters reflect not only the rapid adoption of generative AI tools in marketing but also the growing importance of debates on algorithmic bias, content authenticity and ethical responsibility, which now occupy a central place in

Table 3. Classification of keywords based on co-occurrence analysis

Group	Thematic category	Keyword
Cluster 1 (Red)	Consumer behavior towards Artificial Intelligence	Acceptance, adoption, antecedents, anthropomorphism, authenticity, behavior, chatbots, communication, competence, consequences, consumption, customer satisfaction, customer engagement, customer experience, customer journey, determinants, emotions, engagement, experience, hospitality, impact, information, intention, loyalty, model, online, perceptions, perspective, persuasion, power, privacy, purchase, quality, responses, risk, robots, satisfaction, service, service robots, service quality, technology, technology acceptance model, tourism, trust, user acceptance, value co-creation, warmth
Cluster 2 (Green)	Data analysis and market research with Artificial Intelligence	Advertising, algorithms, artificial neural network, bias, classification, commerce, consumer behavior, data analytics, data mining, decision making, deep learning, forecasting, human, learning, learning systems, machine learning, marketing, meta-analysis, natural language processing, neuromarketing, online advertising, optimization, prediction, recommender systems, sales, sentiment analysis, social networking (online), support vector machine, text analysis, text mining
Cluster 3 (Blue)	Big data and product creation with Artificial Intelligence	Analytics, attitudes, big data, business, challenges, co-creation, consumer, customer relationship, design, ethics, framework, future, innovation, Internet, knowledge, management, marketing strategy, networks, orientation, performance, personalization, product development, psychology, segmentation, strategies, sustainability, systems
Cluster 4 (Yellow)	Artificial Intelligence in Digital Marketing and Social Media	Artificial intelligence, augmented reality, capabilities, ChatGPT, consumer engagement, covid-19, digital marketing, digital transformation, Facebook, gender, generative ai, influencer marketing, Instagram, media, metaverse, retailing, social media, technology adoption, Twitter, virtual influencer, virtual reality
Cluster 5 (Purple)	User-generated online content using Artificial Intelligence	Brand, e-commerce, marketing analytics, online reviews, product, purchase intention, reviews, search, user-generated content, word-of-mouth
Source(s): Own elaboration		

academic and professional discussions (Dwivedi *et al.*, 2023; Grewal *et al.*, 2024). At the same time, the prominence of clusters focused on social presence and influencer marketing signals a research shift toward the experiential and relational dimensions of AI-mediated consumer-brand interactions. This trend is consistent with broader theoretical frameworks on engagement, parasocial interaction and value co-creation, underlining the need to update conceptual models to address these new realities. Beyond identifying dominant keywords and thematic clusters, the co-occurrence analysis reveals a dynamic interplay among research topics such as generative AI, ethics and social presence. The prominence and interconnection of these themes suggest an increasing academic focus on both technological innovation and ethical responsibility in marketing. These findings address RQ2 by showing that the field is evolving to include new debates on transparency, content authenticity and consumer trust issues that are now central to developing AI-driven marketing strategies.

3.2 Reference analysis

The most cited articles correspond to Dwivedi *et al.* (2023) and Dwivedi *et al.* (2021). The former provides a multidisciplinary review of AI's impact, focusing on ChatGPT in education, politics and marketing. The latter centers on social and digital marketing, addressing gaps and future trends related to AI and social media. In third place, Davenport *et al.* (2020) examine the integration of AI into marketing strategies and consumer behavior, analyzing it through three dimensions: technology, operational tasks and robotics. Additionally, Youyou *et al.* (2015) explore the accuracy of computational models in personality judgments. Other notable articles focus on consumer behavior towards AI (Hoyer *et al.*, 2020; Puntoni *et al.*, 2021) and the role of anthropomorphism and robots in purchasing processes (Luo *et al.*, 2019; Belanche *et al.*, 2019, 2020; Chung *et al.*, 2020). Finally, Park and Kim (2022) stand out for their study on the metaverse, analyzing interaction, implementation and applications through three complementary approaches (See Appendix A).

3.3 Author analysis

The authors with the highest number of publications on AI in marketing include Yogesh K. Dwivedi, whose research explores the influence of AI on social media (Dwivedi *et al.*, 2021), consumer behavior in purchasing processes (Wang *et al.*, 2022), the impact of Generative AI (Han *et al.*, 2021) and Big Data applications (Ooi *et al.*, 2023). Sean Sands, in turn, focuses on consumer-AI interactions, particularly addressing how to bridge the gap between consumers and AI (Sands *et al.*, 2020) and AI applications in marketing. His most cited article was co-authored with Campbell (Campbell *et al.*, 2020). Finally, Grewal and Campbell complete the list of leading authors in this field.

3.4 Co-citation analysis

3.4.1 Reference co-citation analysis. The co-citation analysis identifies the most relevant articles based on their joint citation in the literature. Using VOSviewer, a network of 50 articles was generated, each with a minimum of 25 citations, and categorized into four thematic clusters (Figure 4 and Appendix A). It can be observed that Cluster 1 occupies a central position within the network, exhibiting multiple internal connections, which highlights its more significant development and influence. In contrast, the yellow cluster is positioned further from the central core, suggesting that it represents a developing subfield that is not yet fully integrated with the more established research themes.

Cluster 1 (Red): Studies on robots and anthropomorphism. The article by Mende *et al.* (2019) stands out for its analysis of how humanized robots influence consumer interactions and responses in service contexts.

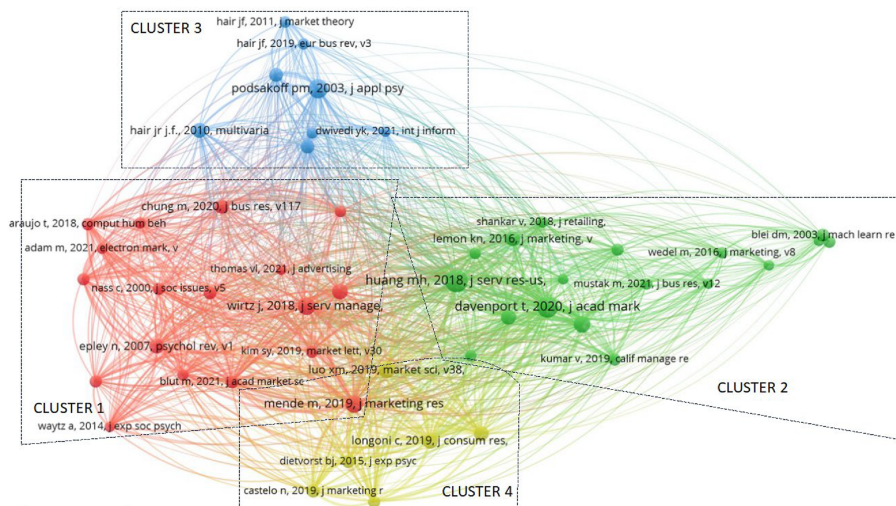


Figure 4. Reference co-citation network. *Source:* Own elaboration (VOSviewer)

Cluster 2 (Green): Focuses on how AI technologies transform marketing strategies, reshaping consumer experiences and decision-making processes. Notably, [Huang and Rust \(2018\)](#) examine the transformation of the service sector through automation, emphasizing the growing importance of human skills such as intuition and empathy.

Cluster 3 (Blue): Addresses methodological and statistical biases in AI research. The study by [Podsakoff et al. \(2003\)](#) identifies four types of biases: sampling, confirmation, measurement and publication.

Cluster 4 (Yellow): Examines algorithm aversion and ethical concerns stemming from a lack of transparency in AI, which can lead to consumer rejection. Among the most influential studies is [Luo et al. \(2019\)](#), which explores the negative effects of chatbot usage in purchasing.

3.4.2 Author co-citation analysis. Continuing with [RQ1](#), the following results highlight the most influential journals and authors, providing a detailed overview of the main contributors who have shaped AI research in marketing during the last two decades. This analysis identifies the most influential authors based on co-citations. A threshold of 25 citations was set, resulting in a network of 290 authors distributed across five clusters ([Figure 5](#)). Cluster 1 (Red): Consisting of 76 authors, this group is led by J. Berger (99 citations, 237 links), whose research focuses on consumer behavior and social influence, and S. Gupta (91 citations, 255 links), specializing in AI and the application of emerging technologies. Cluster 2 (Green): Comprising 69 authors, this cluster highlights J. F. Hair (206 citations, 270 links) and V. Venkatesh (186 citations, 261 links) for their methodological contributions to market research and technological transformation. Cluster 3 (Blue): Containing 66 authors, this group features M. H. Huang (353 citations, 286 links) as the network's most cited and central author. Cluster 4 (Yellow): Encompassing 51 authors, this cluster is led by V. Kumar (231 citations, 283 links) and D. Grewal (148 citations, 277 links), recognized for their research on the interaction between AI and marketing. Cluster 5 (Purple): Consisting of 28 authors, this group is headed by J. Kim (103 citations, 267 links) and S. Sands (70 citations, 234 links), whose work examines the impact of new technologies on consumer behavior and their application in marketing. The graphical distribution indicates that the green, blue, purple and yellow clusters

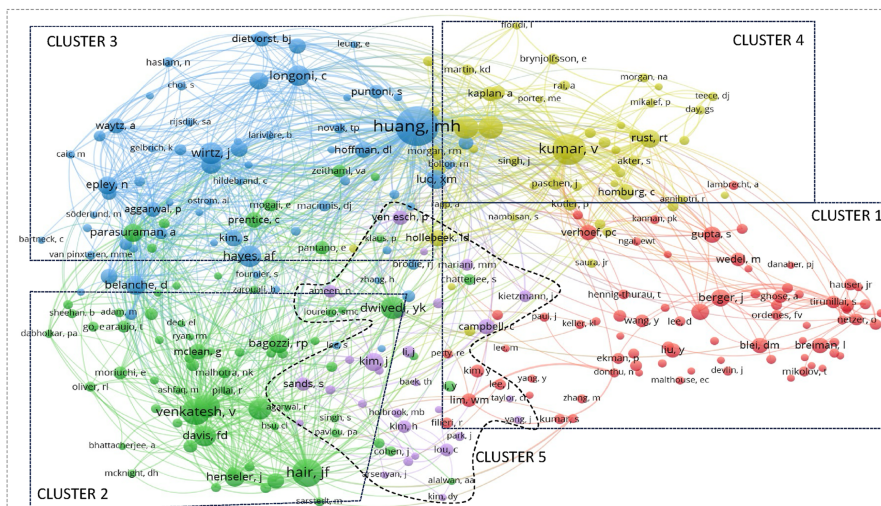


Figure 5. Authors co-citation network. *Source:* Own elaboration (VOSviewer)

are more cohesive and thematically structured. In contrast, the red cluster connects with all other groups, suggesting its interdisciplinary nature.

Analyzing publication trends, thematic clusters and citation networks comprehensively addresses the research questions. It fulfills the overarching objective of updating and extending prior bibliometric reviews in this fast-evolving field. In particular, the co-occurrence analysis goes beyond identifying dominant keywords and clusters by revealing the dynamic interplay between generative AI, ethics and social presence. The prominence and interconnectedness of these themes indicate a growing academic emphasis not only on technological innovation but also on ethical responsibility and trust in AI-driven marketing strategies. This shift is evidenced by the field's increasing engagement with debates around transparency, content authenticity and consumer trust, which are now central to the development and application of AI in marketing.

The emergence of clusters related to generative AI and ethical challenges is closely linked to recent technological advances and growing societal concerns about the reliability and transparency of AI applications. These clusters reflect both the rapid adoption of generative AI tools in marketing and the broader theoretical debates on algorithmic bias, content authenticity and ethical responsibility, which have gained prominence in recent academic and professional discourse (Dwivedi *et al.*, 2023; Grewal *et al.*, 2024). Similarly, the rising importance of social presence and influencer marketing highlights a shift in research priorities toward understanding the experiential and relational dimensions of AI-mediated consumer-brand interactions. This trend aligns with frameworks on engagement, parasocial interaction and value co-creation, underlining the need to update conceptual models and theoretical perspectives. Thus, the clustering of research themes observed in this study directly responds to significant technological and societal changes, pointing to critical areas for further theoretical and practical development.

4. Findings

The following results are presented in direct response to the main objective of this study: to map the intellectual evolution of AI research in marketing between 2000 and 2024, identifying emerging trends, key contributors and knowledge gaps. The review of studies retrieved from Scopus and WOS provides a comprehensive overview of AI research in marketing. Traditionally,

studies have focused on the four core functions of AI: (1) Data capture, (2) Behavior prediction, (3) Task automation and (4) Consumer interaction (Puntoni *et al.*, 2021). One of the most widely discussed topics in the literature is AI anthropomorphism, where humans attribute human-like characteristics to technological agents (Adam *et al.*, 2020). This phenomenon was first described by Mori (1970), who noted that excessive humanization in AI design could lead to discomfort or rejection, a concept known as the uncanny valley theory (Gutiérrez-Aguilar, 2021). Anthropomorphism encompasses physical appearance and functional simulation to elicit natural and trustworthy responses (Li *et al.*, 2016; Van Pinxteren *et al.*, 2019; Abdul-Kader and Woods, 2015). A notable example is the “ELIZA Effect”, which refers to the human tendency to anthropomorphize technological agents (Ekbja, 2008).

Another key issue is privacy concerns and personal data management, which negatively impact AI acceptance due to fears of data misuse and lack of algorithmic transparency (Hasan *et al.*, 2021; Kumar and Suthar, 2023). These factors contribute to distrust and resistance to adoption (Zhang and Curley, 2018). Additionally, recent studies examine user satisfaction with voice assistants (Tassiello *et al.*, 2021; Dellaert *et al.*, 2020) and the motivations driving AI adoption (McLean and Osei-Frimpong, 2019; McLean *et al.*, 2021; Rheu *et al.*, 2020). In summary, AI research in marketing has evolved alongside technological advancements, revealing emerging trends and challenges in key areas (see Figure 6).

The thematic mapping visually represents driving, basic, niche and emerging themes, highlighting areas with high research activity and identifying new or underexplored research frontiers. To answer RQ3, we synthesize the findings from the cluster and thematic analyses, highlighting areas that remain fragmented or insufficiently explored and proposing concrete directions for future investigation.

In Figure 7, the map identifies four thematic categories: driving themes, basic themes, niche themes and emerging or declining themes. Driving themes (upper-right quadrant) are fundamental due to their high relevance and density. Basic themes (lower-right quadrant), while essential because of their high centrality, exhibit lower density and cover broad topics, such as AI-based processing and analysis of large datasets. Moreover, digital marketing, social media and chatbots remain highly relevant in the academic literature. Emerging or declining themes (lower-left quadrant) display low density and centrality, with purchase intention

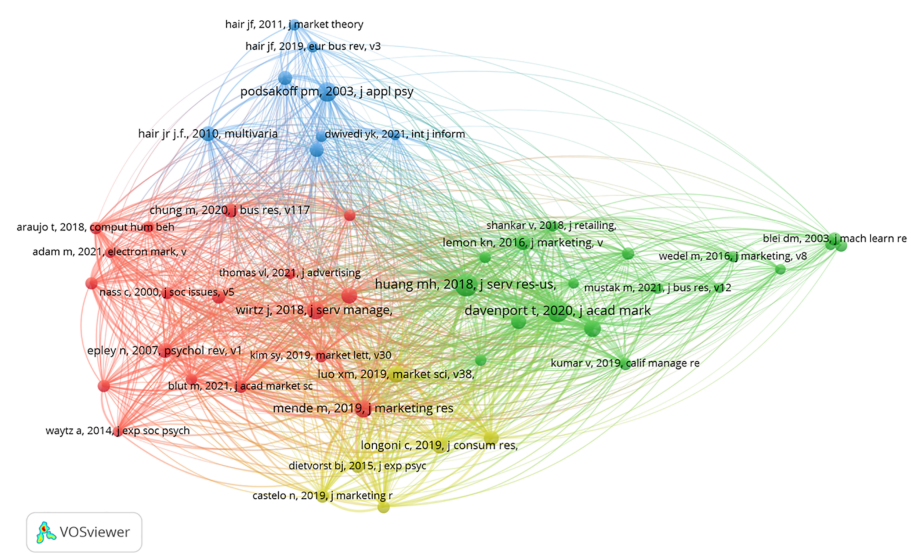


Figure 6. Reference co-citation network. Source: Own elaboration (VOSviewer)

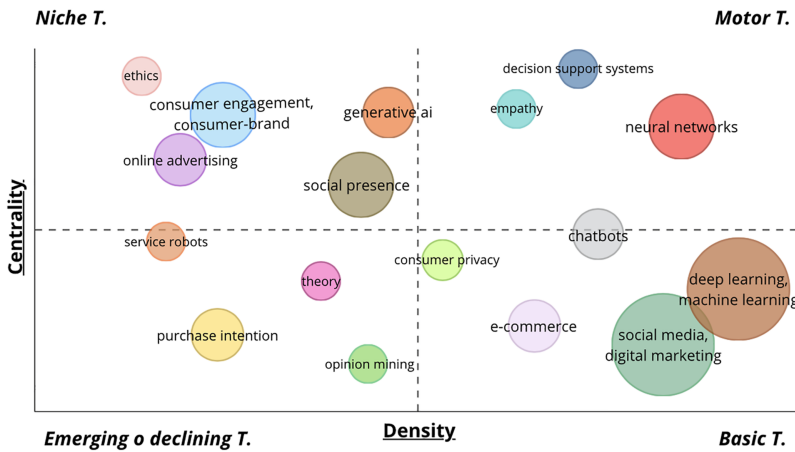


Figure 7. Thematic evolution in scientific research. *Source:* Own elaboration (Biblioshiny)

identified as a declining topic. Despite high density, niche themes (upper-left quadrant) have low centrality, indicating potential for further development. Key topics for more profound AI research include AI ethics, online advertising, Generative AI, consumer engagement and brand relevance.

Figure 8 highlights two significant trends in 2024: Generative AI and ChatGPT. The exponential growth in ChatGPT usage, and consequently Generative AI, has marked a turning point in AI research within marketing. Generative AI facilitates personalized content creation, enhances R&D processes and improves direct customer communication. As a result, its integration is reshaping marketing strategies and business operations (Grewal *et al.*, 2024). Additionally, social media has emerged as an essential marketing tool, maintaining its

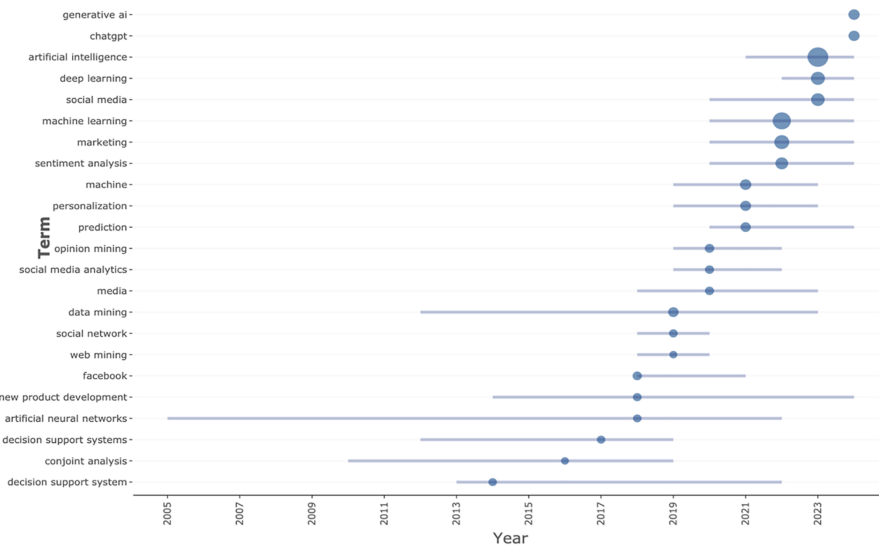


Figure 8. Trending topics (2000–2023). *Source:* Own elaboration (Biblioshiny)

influence, particularly with the rise of influencer marketing. AI enhances audience segmentation, enables emotionally engaging content creation on social platforms and improves campaign impact measurement, maximizing return on investment (ROI).

The inclusion of a comparative summary table further underscores the distinctiveness of this work, illustrating its contribution to expanding current knowledge and clarifying the unique ways in which this study advances the understanding of AI's transformative impact on marketing theory and practice ([Appendix A](#)).

5. Discussion

This study aimed to map the intellectual evolution of AI research in marketing between 2000 and 2024, focusing on emerging trends, key contributors and persistent knowledge gaps. The analysis responds systematically to each research question by integrating bibliometric mapping with thematic and network analyses, ensuring a coherent and critical interpretation of the field's development.

The co-occurrence and author network analyses reveal a substantive shift in research priorities. While earlier work focused primarily on the operational benefits and technological adoption of AI, more recent studies, concentrated in the most interconnected thematic clusters, prioritize issues such as ethical governance, algorithmic transparency and the psychological impact of AI on both consumers and managers. This evolution reflects the necessity for marketing scholarship to expand beyond traditional frameworks and address AI adoption's broader societal and ethical implications.

The empirical findings of this review align with contemporary debates in the literature, particularly the urgency of addressing transparency, trust and social impact in AI-driven marketing ([Dwivedi et al., 2023](#); [Grewal et al., 2024](#)). Foundational models such as the Technology Acceptance Model (TAM) and UTAUT, long used to explain adoption via perceived usefulness and ease of use, are increasingly supplemented by new considerations, including algorithmic bias, content authenticity and regulatory concerns.

From a theoretical perspective, these findings call for the revision and extension of established frameworks to encompass the complexity of contemporary AI applications in marketing. Identifying emergent clusters, such as AI-driven influencer marketing and social presence, indicates a need for renewed theoretical inquiry into human-machine relationships, authenticity and brand engagement in digital contexts. There is also a growing demand for integrative models incorporating ethical, psychological and societal dimensions.

Concerning [RQ1](#), the analysis documents an acceleration in publication output from 2017 onwards, highlights the most influential journals and authors, and provides a detailed mapping of international contributions. For [RQ2](#), five major thematic clusters are identified, encompassing areas such as consumer behavior, big data and AI in social media, as well as new domains such as generative AI and ethical challenges. [RQ3](#) is addressed by identifying persistent gaps, most notably algorithmic bias, content authenticity and the regulatory challenges associated with generative AI, that represent critical priorities for future research.

The prominence of generative AI and ethical concerns reflects a fundamental shift in the field, highlighting the need to update existing theoretical frameworks and incorporate new perspectives on human-machine interaction and ethical responsibility. Methodologically, this study advances prior reviews by adopting a systematic, high-quality sampling approach and employing tools such as Bibliometrix and VOSviewer. This design ensures a comprehensive and reliable field mapping while highlighting the limitations inherent to bibliometric analysis, essentially the challenges of capturing qualitative depth and excluding specific literature sources not indexed in major databases.

The findings of this bibliometric review provide several actionable insights for marketing practitioners facing the rapid evolution of AI. First, the emergence of generative AI as a prominent research cluster underscores the urgency for organizations to integrate these technologies strategically in their marketing operations. Generative AI now enables the

development of highly personalized content, real-time customer interactions and automated campaign management, allowing firms to enhance efficiency and consumer engagement.

Ethical governance emerges as another key managerial consideration. The increasing focus on algorithmic transparency and bias in the literature highlights the growing scrutiny organizations face from regulators and consumers. Managers should establish robust frameworks for ethical oversight, including creating multidisciplinary teams to oversee AI deployment, routine auditing of algorithms for fairness and transparency and proactive communication with stakeholders regarding data usage and decision-making processes.

Moreover, the rise of social presence and influencer marketing as critical research themes point to firms' need to invest in cultivating authentic brand personas in digital environments. AI-powered tools can assist in identifying relevant influencers, tailoring messages for specific audience segments and monitoring engagement metrics. However, this technological sophistication must be balanced with authenticity, ensuring that influencer partnerships and AI-generated content foster genuine connections rather than eroding trust through perceived artificiality or manipulation.

Another important implication is the growing requirement for upskilling and organizational learning. As AI technologies advance, marketing teams must be equipped with technical and ethical competencies. Continuous professional development programs, cross-functional training and collaboration with external experts or academic partners are recommended to ensure personnel are prepared to harness AI's full potential, adapt to evolving best practices and respond swiftly to emerging risks or opportunities. Finally, the study highlights the necessity for marketing leaders to adopt a proactive, rather than reactive, stance toward AI integration. This means tracking technological trends and anticipating shifts in consumer expectations, regulatory landscapes and societal attitudes toward AI.

This study provides a longitudinal perspective on the use of AI in marketing, but in recent years, academic interest has shifted toward Generative AI. Its adoption has accelerated with the development of large language models (LLMs) such as ChatGPT-5, Gemini 2.5, Claude 3.5, Copilot, Grok and open-source models like LLaMA, Mistral or Deepseek (Maslej *et al.*, 2025). These advances have consolidated Generative AI as a tool capable of creating innovative solutions in business management (Raisch and Fomina, 2024) and enhancing workforce skills (Brynjolfsson *et al.*, 2025). In marketing, the debate no longer focuses on adoption but on the challenges of implementation, accountability and autonomy, with the hybrid human-AI model emerging as the most effective approach (Arora *et al.*, 2025). However, the need for constant human oversight contrasts with the rapid corporate integration of AI, where a significant gap persists between the recognition of risks and the implementation of Responsible AI measures (Maslej *et al.*, 2025). Added to this are the rise in AI-related incidents, the persistence of implicit biases and increasing public distrust. Finally, autonomous LLM-based agents open up new opportunities but also intensify debates on security, malicious use and resilience against adversarial attacks (Xi *et al.*, 2025).

This study presents limitations inherent to bibliometric analyses. While it offers a quantitative and panoramic view of the evolution of AI research in marketing, its approach does not allow for a deep qualitative understanding of complex phenomena. Furthermore, the exclusive use of Scopus and Web of Science may have excluded relevant literature from other repositories, applied research not indexed in traditional outlets and recent contributions published in books, book chapters or other non-conventional media.

Several priority directions for future research are delineated based on the main gaps and thematic clusters identified in this analysis. First, it is essential to conduct empirical studies that explore the real effects of generative AI on brand communication, consumer perception and content authenticity. Likewise, methodological advances are needed to detect, explain and mitigate algorithmic biases in marketing systems, with particular attention to high-impact sectors and vulnerable groups. Future studies should combine quantitative bibliometric techniques with qualitative approaches to enrich understanding of AI's real-world implications. In addition, further research is warranted on the ethical and regulatory

implications of AI-driven marketing, especially in light of evolving policies and international standards. Finally, the role of AI in generating social presence and enhancing consumer engagement represents a promising area that requires more empirical investigation to clarify its effects on trust, long-term relationships and underlying psychological processes.

6. Conclusions

This study delivers the most current and systematic bibliometric mapping of AI in marketing research from 2000 to 2024. It documents the field's intellectual development and identifies its principal contributors and enduring research gaps. The exponential growth in AI-related publications since 2017 demonstrates the rapid adoption of these technologies within marketing, with the United States, China, India and the United Kingdom shaping global research trends. Analysis revealed five main thematic clusters, from consumer behavior and market research to big data, digital marketing and user-generated content, highlighting the growing relevance of technological innovation and the ethical, social and psychological challenges AI poses. Co-citation analyses identified four dominant research streams: the application of AI in data analysis, the transformation of marketing strategies, methodological advancements and the exploration of ethical and informational challenges. Traditionally, AI anthropomorphism has been one of the most recurrent themes in the literature on AI and marketing. However, the most recent and relevant studies show a significant shift in this trend, moving toward new lines of research that range from the integration of AI into social and digital marketing to the analysis of the impact of models such as ChatGPT in different domains. In this context, generative AI is currently consolidating its position at the core of research in this field. Therefore, the rising prominence of generative AI concerns over content authenticity and algorithmic bias reflects a field in transition, with implications that extend well beyond operational improvements. This work advances the literature by offering integrated findings and a research agenda attuned to AI's evolving opportunities and challenges in marketing, providing actionable insights for academics and practitioners. In addition to offering an updated and systematic mapping of AI research in marketing, this study provides a foundation that extends beyond descriptive purposes. The results contribute to the consolidation of theoretical frameworks by highlighting critical issues such as algorithmic bias, content authenticity and consumer trust in the use of generative AI. From a managerial perspective, the identification of emerging themes offers insights for integrating generative AI and related technologies into marketing strategies in a transparent and ethical manner. Furthermore, the study delineates clear avenues for future research, including empirical investigations on the impact of generative AI on consumer engagement, methodological developments to address biases and the examination of regulatory and governance frameworks.

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Supplementary material

The supplementary material for this article can be found online.

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